Synchronous Circuits :-

* memory must flip-flop or array of flip-flops (register)
* If register, then all FF must be driven by same clock.

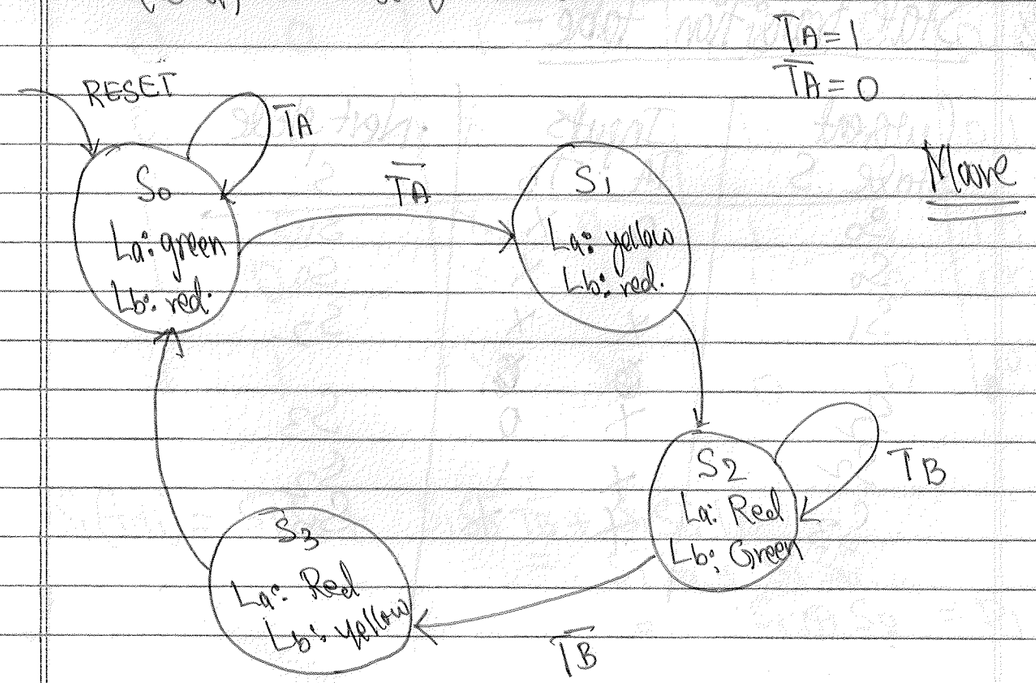
29/8/26

UNIT 2: DIGITAL BUILDING BLOCKS :-

no. of registers : K
no. of possible states : 2^K

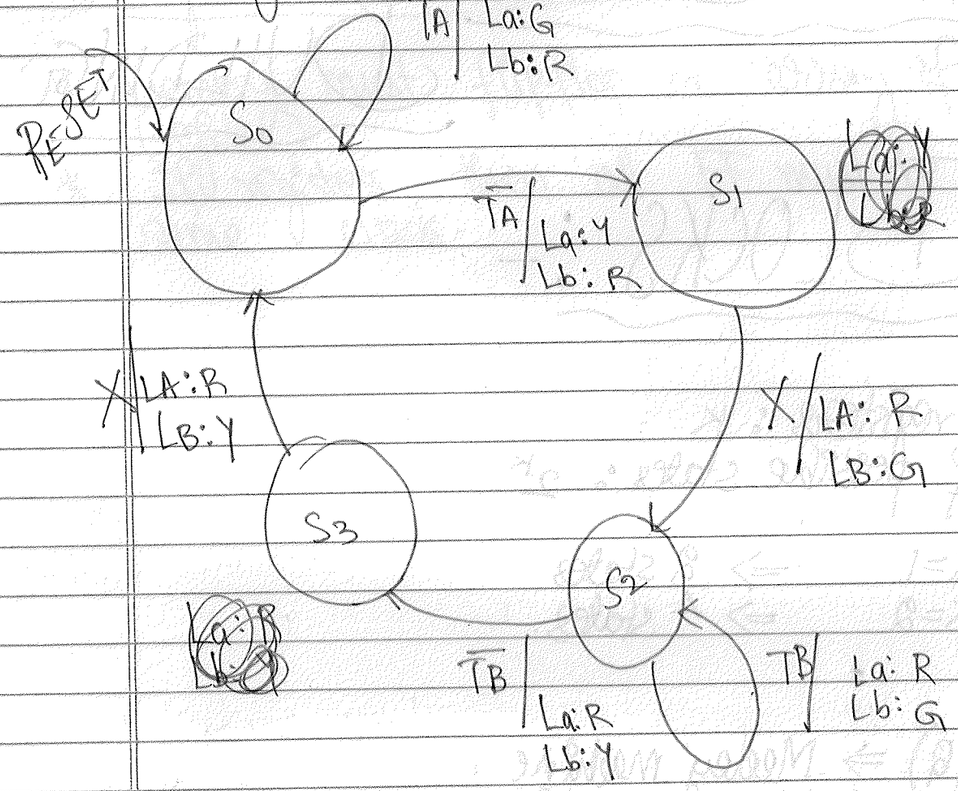
K=1 => 2 states
K=2 => 4 states.

Y(X,Q) => Meeley machine
Y(Q) => Moore machine.



Achiever
Date: / /

Mealy –



Moore State transition table –

Current State S	Inputs T_A	Inputs T_B	Next state S'
S_0	0	X	S_1
S_0	1	X	S_0
S_1	X	X	S_2
S_2	X	0	S_3
S_2	X	1	S_2
S_3	X	X	S_0

State Encoding –

S_0	00	binary assignment
S_1	01	
S_2	10	
S_3	11	

Output encoding –

green	00
yellow	01
red	10

State Transition Table

Moore

Current State S_1	Current State S_0	Inputs T_A	Inputs T_B	Next State S_1	Next State S_0	Output L_{A1}	L_{A0}	L_{B1}	L_{B0}
0	0	0	X	0	1	0	0	1	0
0	0	1	X	0	0	0	0	1	0
0	1	X	X	1	0	0	1	1	0
1	0	X	0	1	1	1	0	0	0
1	0	X	1	1	0	1	0	0	0
1	1	X	X	0	0	1	0	0	1

$$S_1(t+1) = \bar{S_1}S_0 + S_1\bar{S_0}\,\bar{T_B} + S_1\bar{S_0}T_B$$
$$= \bar{S_1}S_0 + S_1\bar{S_0}$$
$$= S_1 \oplus S_0 = P_1$$

$S_0(t+1) = S_1\bar{S_0}\bar{T_A} + S_1\bar{S_0}T_B = D_2$

$L_{A1} = S_1\bar{S_0} + S_1S_0$

$L_{A0} = S_1\bar{S_0}$

$L_{B1} = S_1\bar{S_0} + \bar{S_1}\bar{S_0}$

$L_{B0} = S_1S_0$

Mealey

State Transition Table –

Current State		Input		Next state		Outputs			
S_1	S_0	T_A	T_B	S_1	S_0	L_{A1}	L_{A0}	L_{B1}	L_{B0}
0	0	0	X	0	1	0	1	1	0
0	0	1	X	0	0	0	0	1	0
0	1	X	X	1	0	1	0	0	0
1	0	X	0	1	1	1	0	0	1
1	0	X	1	1	0	1	0	0	0
1	1	X	X	0	0	1	0	0	1

$L_{A1} = S_1S_0 + S_1\bar{S_0} + S_1S_0$

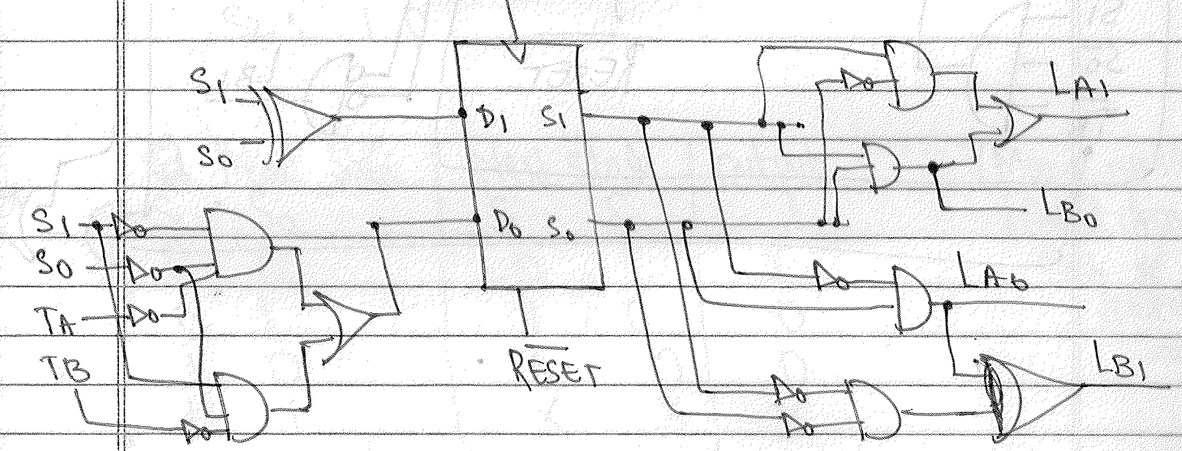
$L_{A0} = \bar{S_1}\bar{S_0}\bar{T_A}$

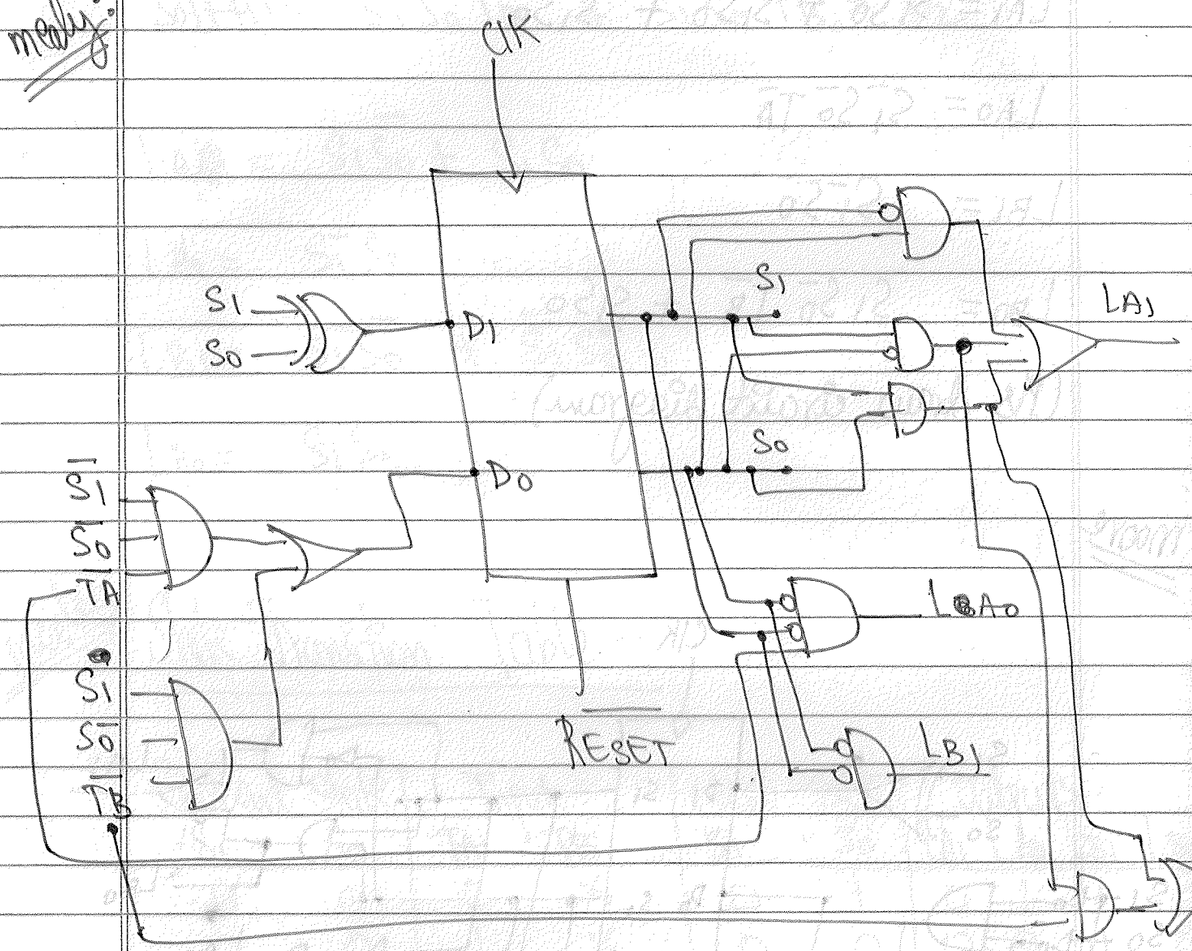
$L_{B1} = \bar{S_1}\bar{S_0}$

$L_{B0} = S_1\bar{S_0}\bar{T_B} + S_1S_0$

(Pls draw circuit diagram)

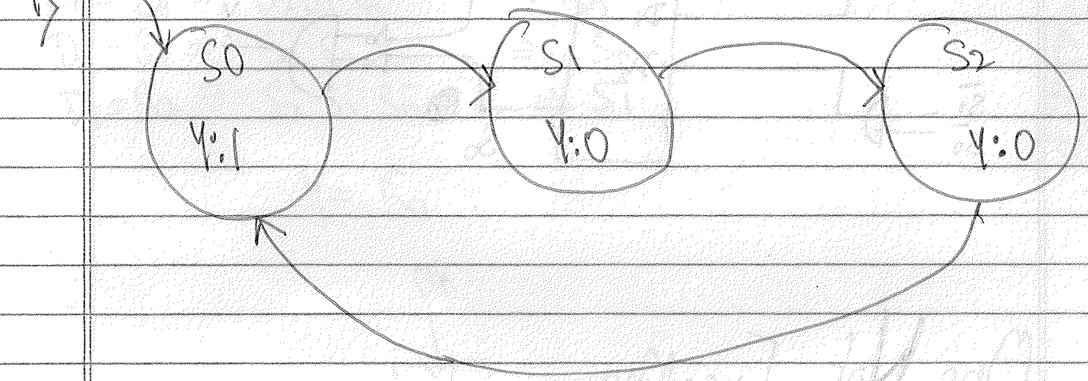
Moore





Question:

reset.



Soln:

S0 : 0 0
S1 : 0 1
S2 : 1 0

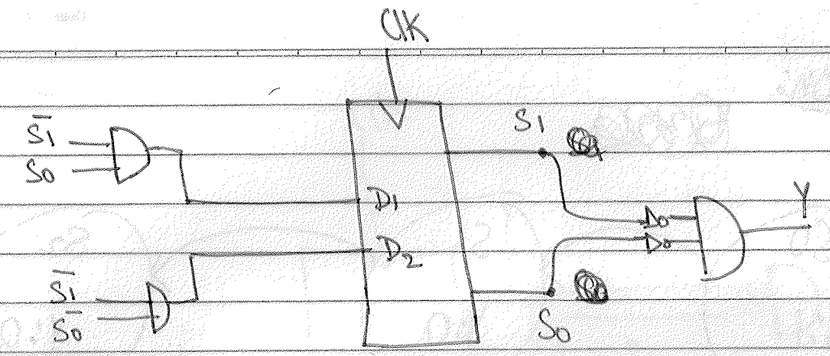
Present State		Next state		output
S_1	S_0	S_1	S_0	
0	0	0	1	1
0	1	1	0	0
1	0	0	0	0
1	1	X	X	X

(Moore machine (Output depends only on present state)

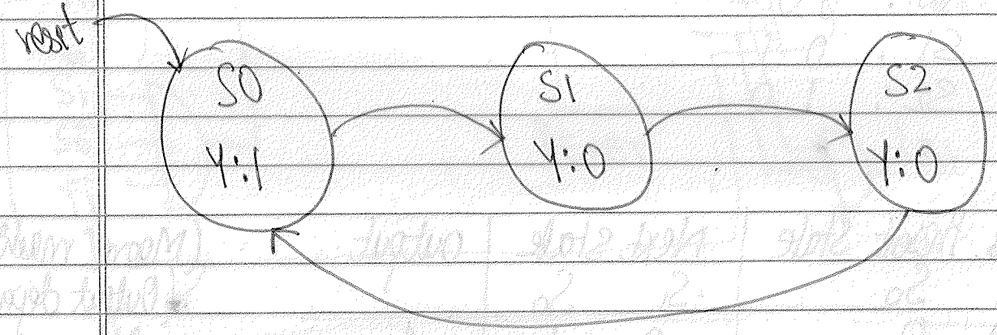
$S_1 = \overline{S_1} S_0 = D_1$

$S_0 = \overline{S_1}\,\overline{S_0} = D_2$

$Y = \overline{S_1}\,\overline{S_0}$



One-Hot Encoding –



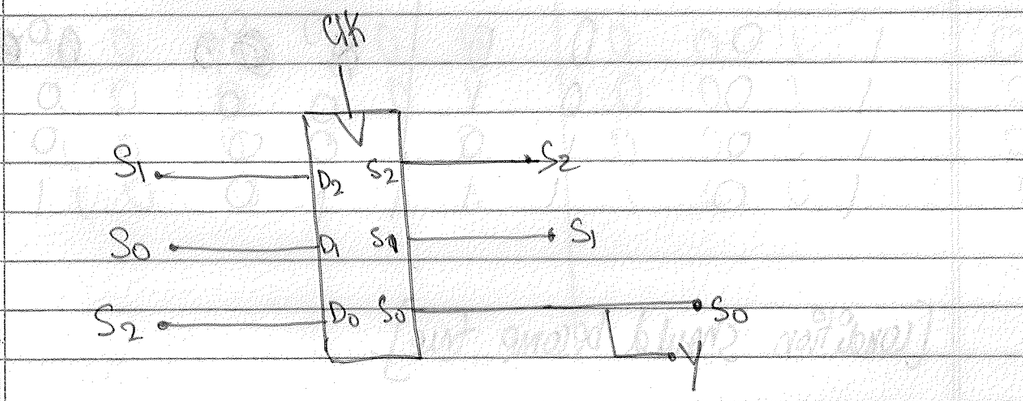
S_0: 0 0 1
S_1: 0 1 0
S_2: 1 0 0

n states ⇒ n bits

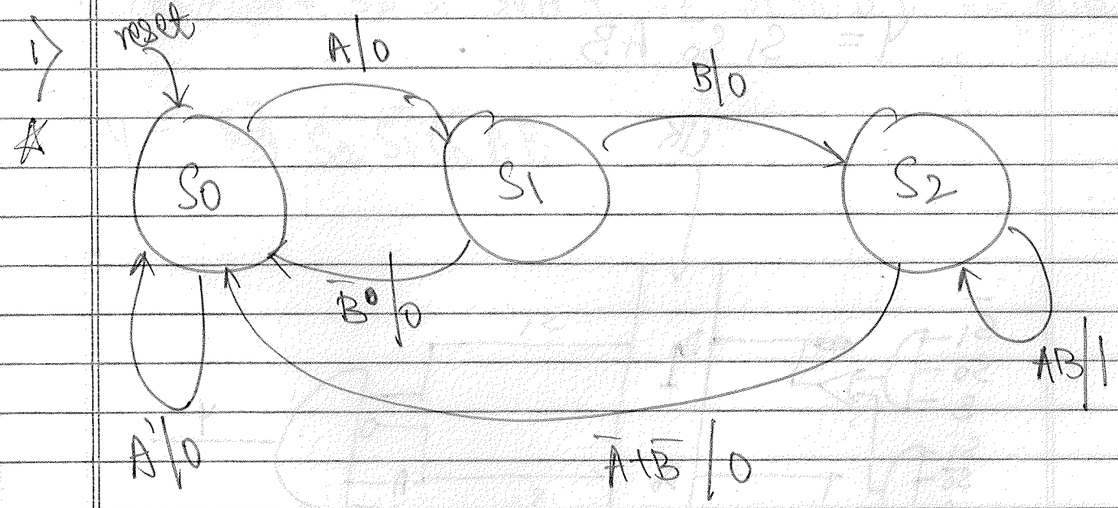
Present State			Next State			Output
S_2	S_1	S_0	S_2	S_1	S_0	Y
0	0	1	0	1	0	1
0	1	0	1	0	0	0
1	0	0	0	0	1	0

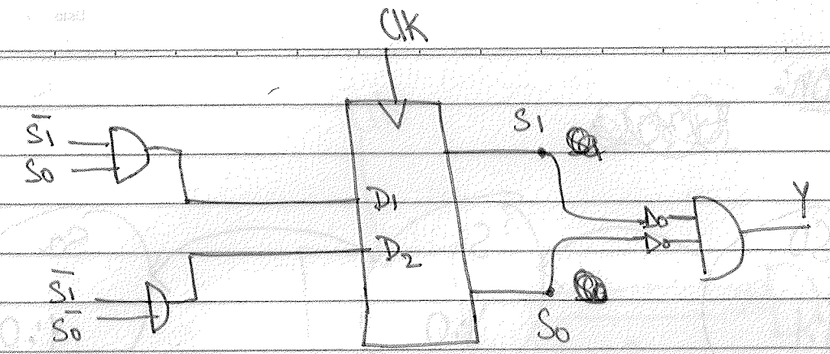
$D_0 = S_0 = S_2 \bar{S_1} S_0 \Rightarrow S_2$
$D_1 = S_1 = \bar{S_2} \bar{S_1} S_0 \Rightarrow S_0$
$D_2 = S_2 = \bar{S_2} S_1 \bar{S_0} \Rightarrow S_1$

$Y = S_0$.

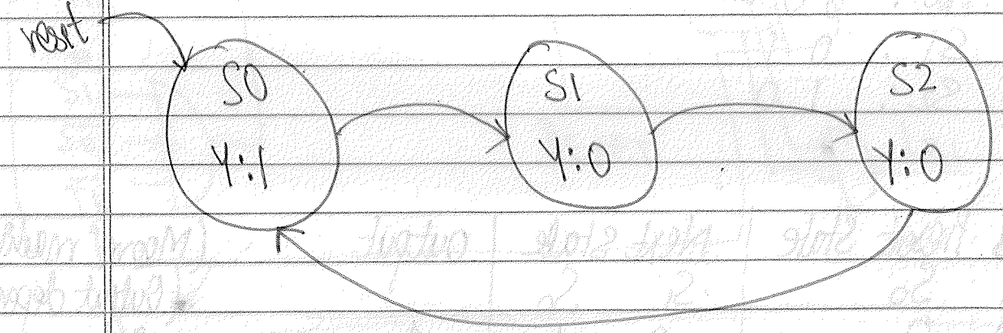


Question –





One-Hot Encoding –



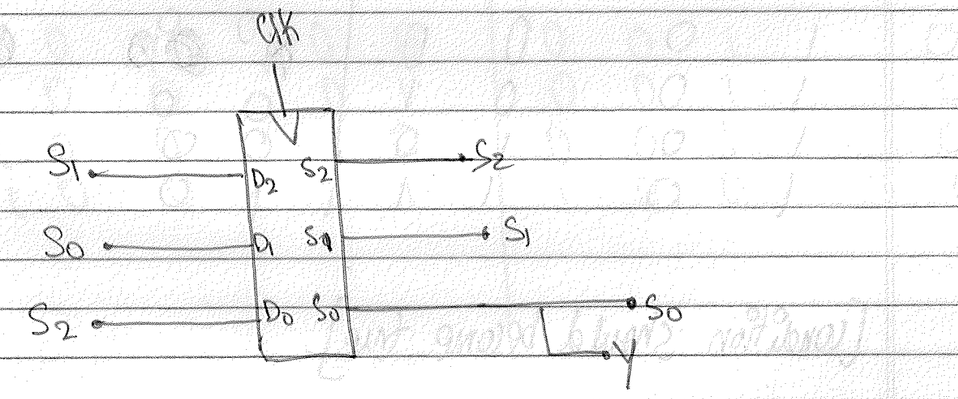
S0: 0 0 1
S1: 0 1 0
S2: 1 0 0

n states ⇒ n bits

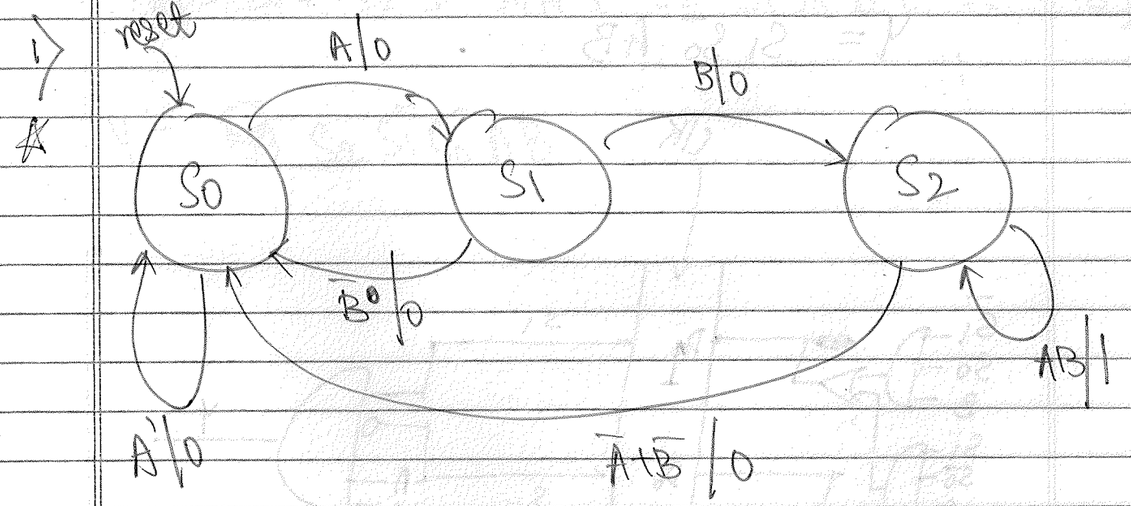
Present State			Next State			Output
S_2	S_1	S_0	S_2	S_1	S_0	Y
0	0	1	0	1	0	1
0	1	0	0	0	1	0
1	0	0	0	0	1	0

$D_0 = S_0 = S_2 \bar{S_1} S_0 \Rightarrow S_2$
$D_1 = S_1 = \bar{S_2} S_1 \bar{S_0} \Rightarrow S_0$
$D_2 = S_2 = \bar{S_2} S_1 \bar{S_0} \Rightarrow S_1$

$Y = S_0$.



Question –



Binary

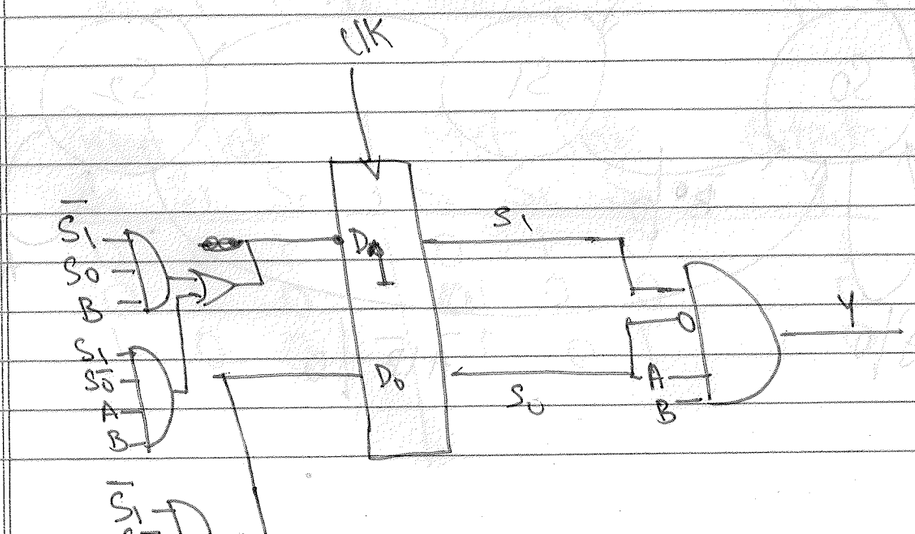
Present State S_1	S_0	Inputs A	B	Next State S_1	S_0	Output Y
0	0	1	X	0	1	0
0	0	0	X	0	0	0
0	1	X	0	0	0	0
0	1	X	1	1	0	0
1	0	0	0	0	0	0
1	0	0	1	0	0	0
1	0	1	0	0	0	0
1	0	1	1	1	0	1

[condition should become true]

$S_1 = \bar{S_1} S_0 B + S_1 \bar{S_0} AB = D_1$

$S_0 = \bar{S_1} \bar{S_0} A = D_0$

$Y = S_1 \bar{S_0} AB$



One Hot

Present State S_2	S_1	S_0	Inputs A	B	Next State S_2	S_1	S_0	Output Y
0	0	1	0	X	0	0	1	0
0	0	1	1	X	0	1	0	0
0	1	0	X	0	0	0	1	0
0	1	0	X	1	1	0	0	0
1	0	0	0	0	0	0	1	0
1	0	0	0	1	0	0	1	0
1	0	0	1	0	0	0	1	0
1	0	0	1	1	1	0	0	1

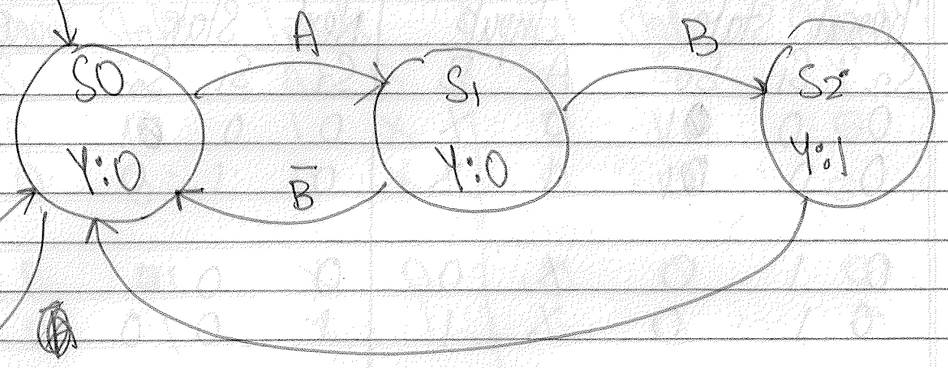
$D_2 = S_2 = S_2 \bar{S_1} \bar{S_0} B + S_2 \bar{S_1} \bar{S_0} AB$

$D_1 = S_1 = \bar{S_2} \bar{S_1} S_0 A$

$D_0 = S_0 = \bar{S_2} \bar{S_1} S_0 \bar{A} + \bar{S_2} S_1 \bar{S_0} \bar{B} + S_2 (\bar{A} + \bar{B})$

$Y = S_2 \bar{S_1} \bar{S_0} AB$

2)



Soln:

Binary Encoding

Present state S_1	Present state S_0	Input A	Input B	Next State S_1	Next State S_0	Output Y
0	0	0	X	0	0	0
0	0	1	X	0	1	0
0	1	X	0	0	0	0
0	1	X	1	1	0	0
1	0	X	X	0	0	1

$Y = S_1 \bar{S_0}$

$D_1 = \bar{S_1} S_0 B$

$D_0 = \bar{S_1} \bar{S_0} A$

One hot encoding

Present state S_2	S_1	S_0	Input A	Input B	Next State S_2	S_1	S_0	Output Y
0	0	1	0	X	0	0	1	0
0	0	1	1	X	0	1	0	0
0	1	0	X	0	0	0	1	0
0	1	0	X	1	1	0	0	0
1	0	0	X	X	0	0	1	1

$Y = S_2 \bar{S_1} \bar{S_0}$

$D_2 = \bar{S_2} S_1 \bar{S_0} B$

$D_1 = \bar{S_2} \bar{S_1} S_0 A$

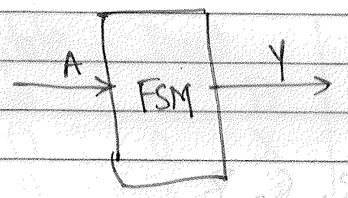
$D_0 = \bar{S_2} \bar{S_1} \bar{S_0} \bar{A} + \bar{S_2} S_1 \bar{S_0} \bar{B} + S_2 \bar{S_1} \bar{S_0}$

Question- (slide 157)

1) P Hacker owns a pet robotic snail with an FSM. The snail crows from Left to right along a paper tail containing a sequence of 0's & 1's. On each clock cycle, the snail crawls to the next bit. The snail smiles when the last 2 bits it has crawled over R from left to right 01.

Design FSM to compute when the snail should smile. The input A is the bit underneath the snails antinae. The output Y is true when the snail smiles.

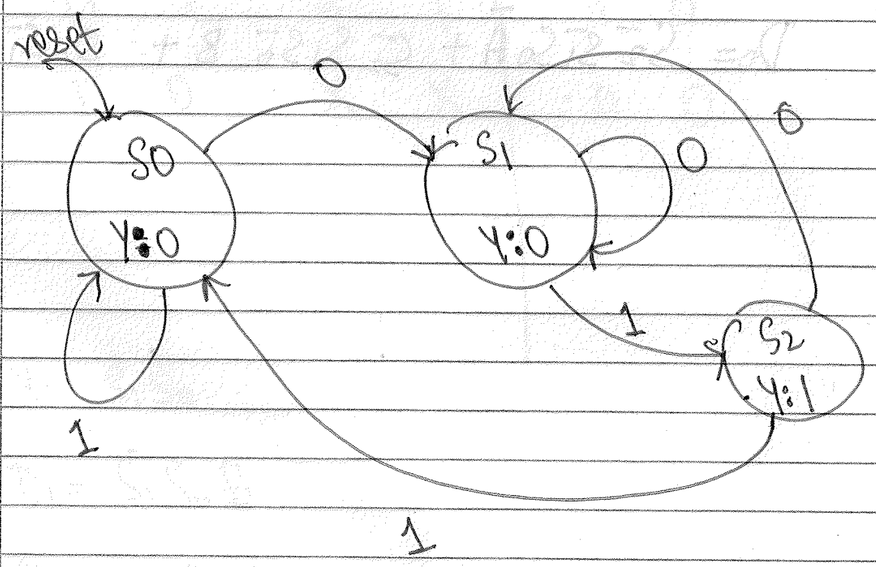
Soln:



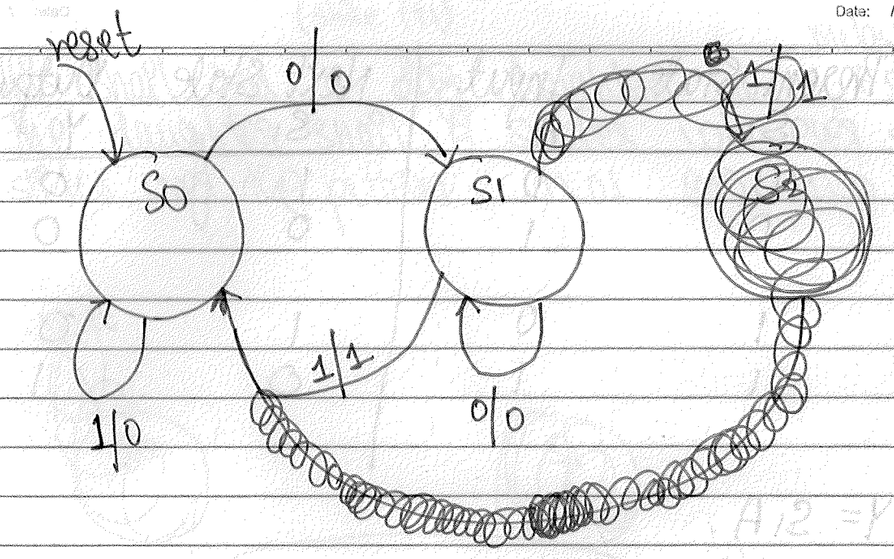
01 Y=1

--- Y=0

Moore



Mealy



Moore ⇒ State Transition Table.

Present State S_1	Present State S_0	Input A	Next State S_1	Next State S_0	Output Y
0	0	0	0	1	0
0	0	1	0	0	0
0	1	0	0	1	0
0	1	1	1	0	0
1	0	0	0	1	1
1	0	1	0	0	1

$Y = S_1 \bar{S_0}$

$S_0 = D_0 = \bar{S_1} \bar{S_0} \bar{A} + \bar{S_1} S_0 \bar{A} + S_1 \bar{S_0} A$

$S_1 = D_1 = \bar{S_1} S_0 A$

melay

Present State S_1	Input A	Next State S_1	Output Y
0	0	1	0
0	1	0	0
1	0	1	0
1	1	0	1

$Y = S_1 A$

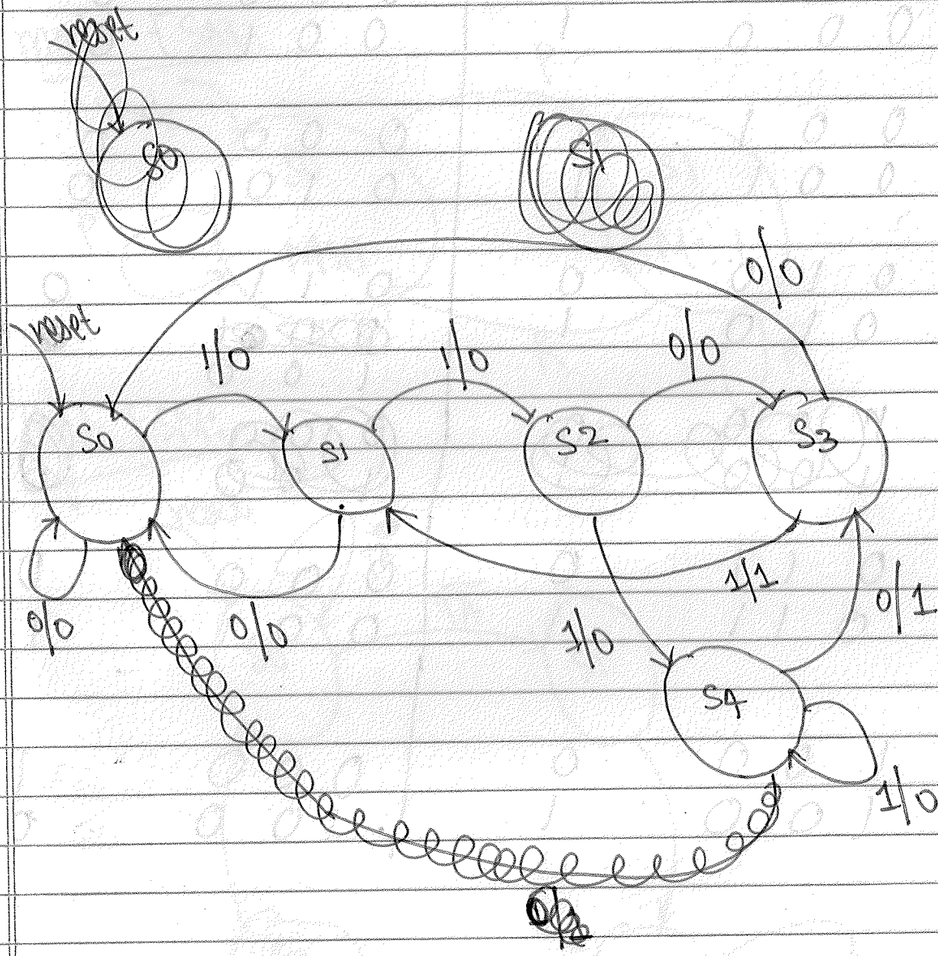
$S_1 = D_1 = S_1\bar{A} + S_1\bar{A} = \bar{A}$

(slide 191)

melay.

2. This snail has a dog daughter has a FSM brain. The daughter snail ☺ smiles whenever she slides over the pattern 1101 (or) 1110

Soln:

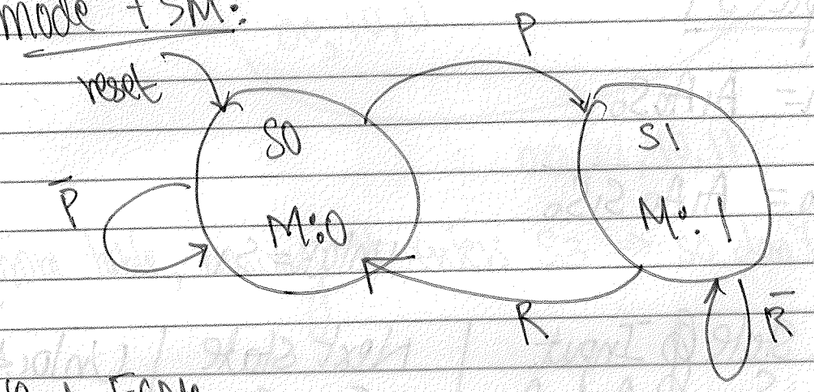


	Present State S_2 S_1 S_0	Input A	Next State S_2 S_1 S_0	Output Y
S0	0 0 0	0	0 0 0	0
	0 0 0	1	0 0 1	0
S1	0 0 1	0	0 0 0	0
	0 0 1	1	0 1 0	0
S2	0 1 0	0	0 1 1	0
	0 1 0	1	1 0 0	0
S3	0 1 1	0	0 0 0	0
	0 1 1	1	0 0 1	1
S4	1 0 0	0	0 0 0	1
	1 0 0	1	1 0 0	0

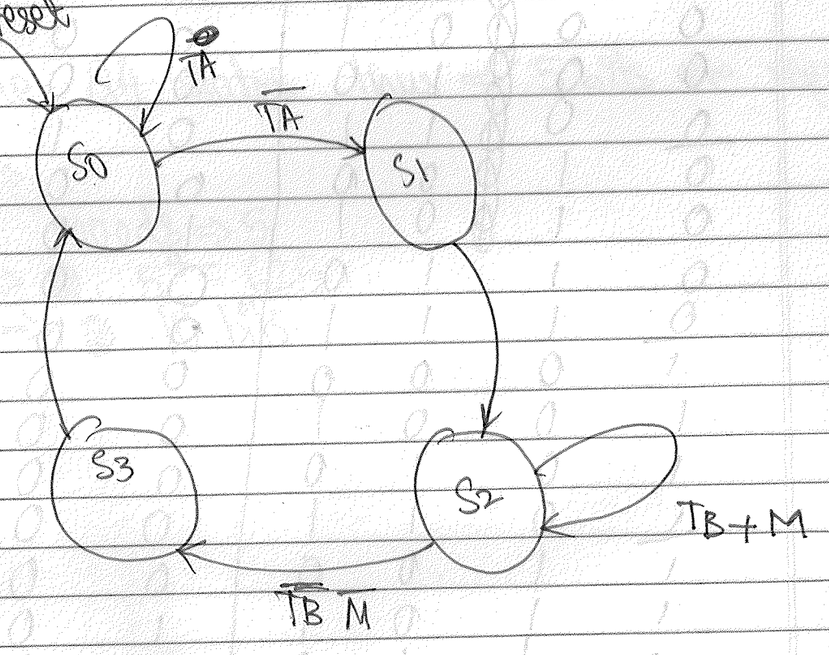
Factoring of State Machines —

Ex:

mode FSM:



Input FSM:



→ check for combinational output

→ Input + present state ⇒ mealy

→ present state ⇒ more.

→ no combinational output

→ moore machine.

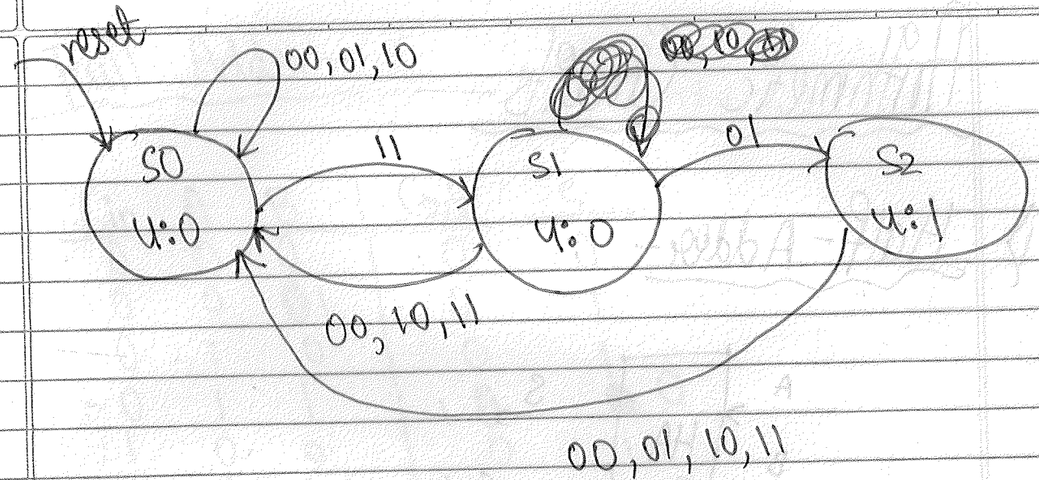
example 3.9

$S_1 = D_1 = \bar{A_1} A_0 S_0$

$S_0 = D_0 = A_1 A_0 \bar{S_1} \bar{S_0}$

Unlock = S_1

Present state		Input		Next state		Unlock
S_1	S_0	A_1	A_0	S_1	S_0	
0	0	0	0	0	0	0
0	0	0	1	0	0	0
0	0	1	0	0	0	0
0	0	1	1	0	1	0
0	1	0	0	0	0	0
0	1	0	1	1	0	0
0	1	1	0	0	0	0
0	1	1	1	0	0	0
1	0	0	0	0	0	1
1	0	0	1	0	0	1
1	0	1	0	0	0	1
1	0	1	1	0	0	1
1	1	0	0	0	0	1
1	1	0	1	1	0	1
1	1	1	0	0	0	1
1	1	1	1	0	0	1



(From table, no way it enters S_3 ⇒ so don't write)

Ex 3.19 ⇒

There are 24 states, thus ⇒ 5 bits are required.

Ex 3.20 ⇒

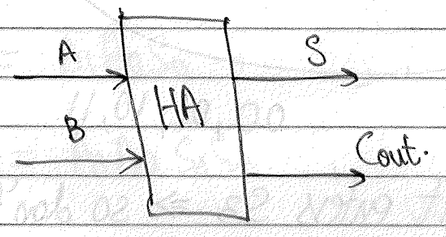
moods ⇒ 5

States ⇒ $5^4 = 625$

bits ⇒ 10 bits.

Arithmetic Circuits —

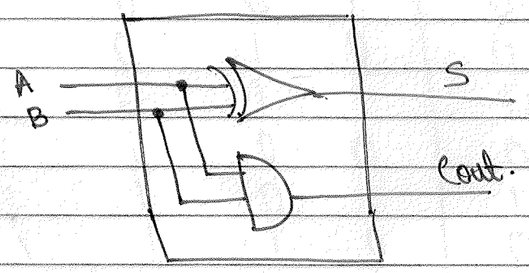
1) Half-Adder —



A	B	S	Cout
0	0	0	0
0	1	1	0
1	0	1	0
1	1	0	1

$S = A \oplus B$

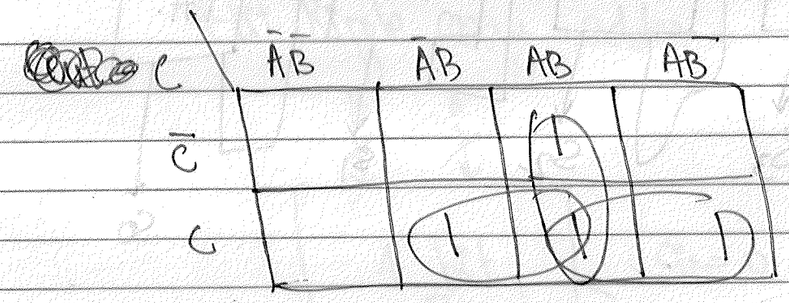
$Cout = AB$



1 bit HA

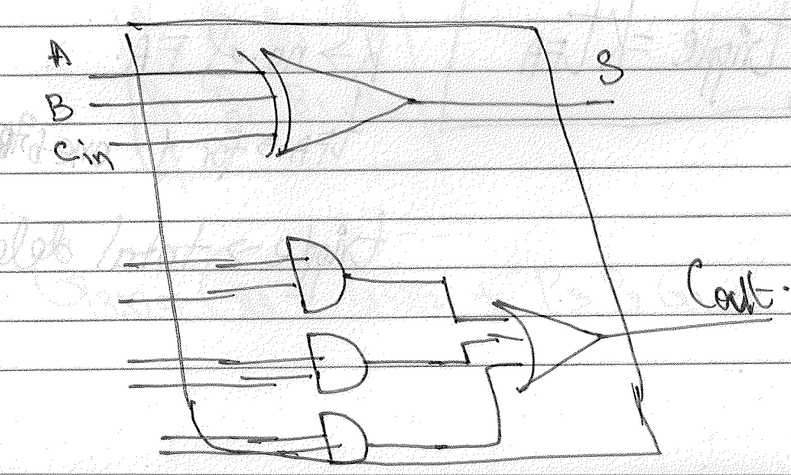
2) Full Adder —

Cin	A	B	Cout	S
0	0	0	0	0
0	0	1	0	1
0	1	0	0	1
0	1	1	1	0
1	0	0	0	1
1	0	1	1	0
1	1	0	1	0
1	1	1	1	1



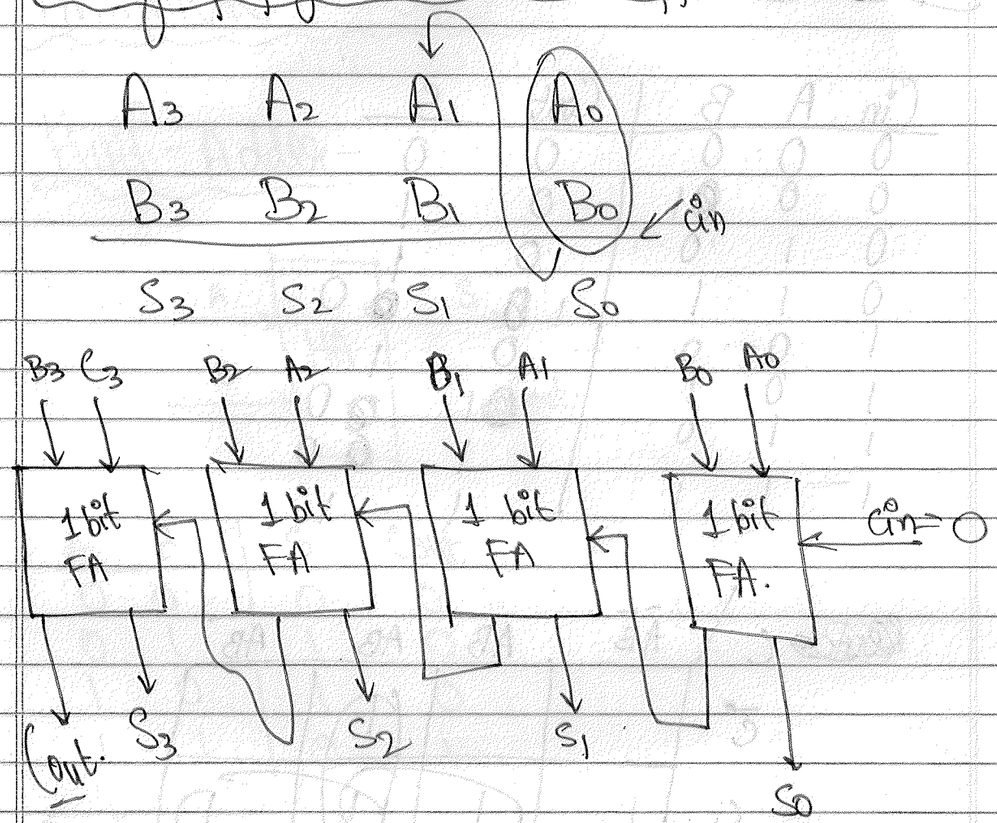
$Cout = AB + CinB + CinA$

$S = A \oplus B \oplus C$



1 bit FA.

i) Carry propogate Adder (Ripple adder) —



2 → n bit number
result → (n+1) bit.

$$\text{Tripple} = K t_{FA}$$

K → no. of FA.

t_{FA} → for 1 one-bit FA

tripple → total delay.

ii) Carry lookahead Adder —

G → generate. P → propogate.

$G_i = A_i B_i$

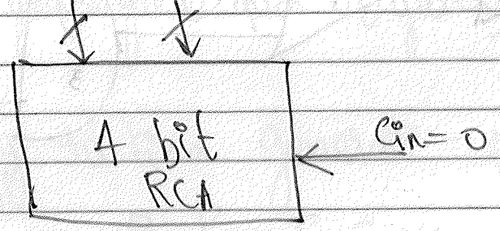
$P_i = A_i \oplus B_i$

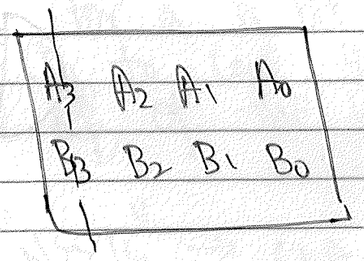
ith → column

$\langle C_i = G_i + P_i C_{i-1} \rangle$

16 bit Adder:

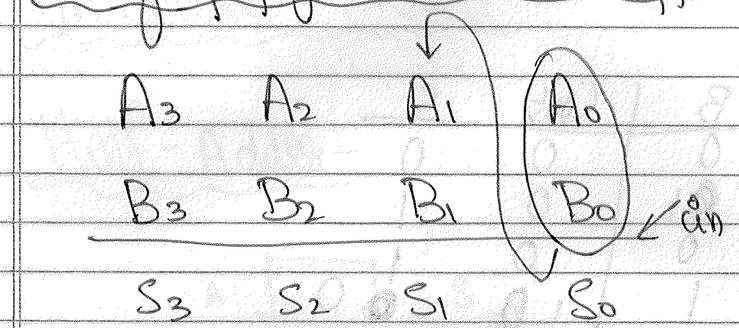
4-bit Ripple carry adder.

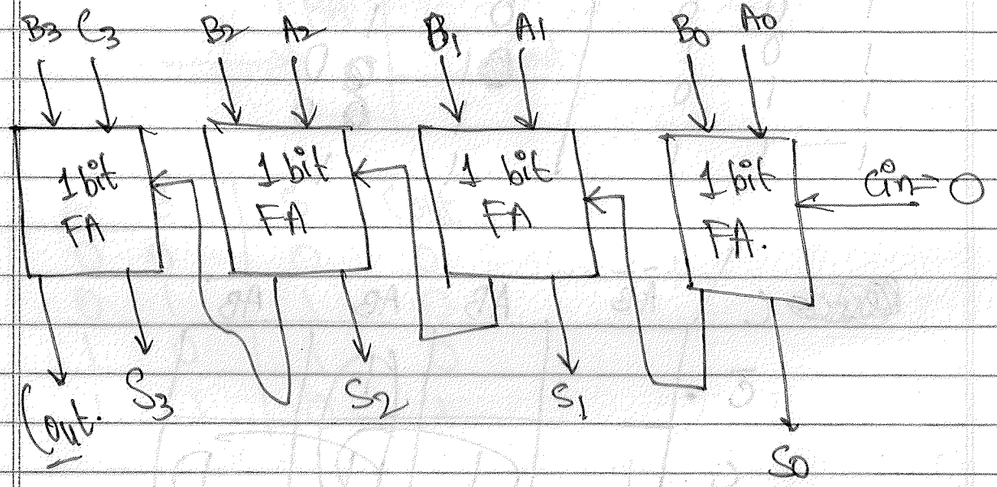




$G_{3:0} = G_3 + P_3 G_2 + P_3 P_2 G_1 + P_3 P_2 P_1 G_0$

i) Carry propagate Adder (Ripple adder) —





2 → n-bit Number
result → (n+1) bit.

%

$t_{ripple} = k t_{FA}$

K → no. of FA.

t_{FA} → for 1 one-bit FA

ripple → total delay.

ii) Carry lookahead Adder —

G → generate. P → propagate.

$G_i = A_i B_i$

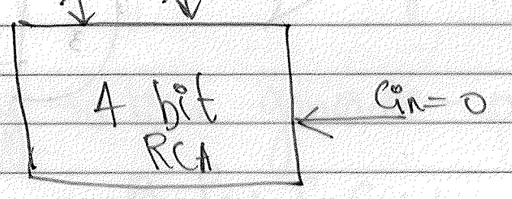
$P_i = A_i \oplus B_i$

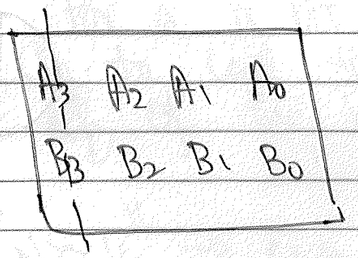
ith → column

$\langle C_i = G_i + P_i C_{i-1} \rangle$

16 bit Adder:

4-bit Ripple carry adder.





$G_{3:0} = G_3 + P_3 G_2 + P_3 P_2 G_1 + P_3 P_2 P_1 G_0$